

Applied A.I. Solutions

Foundations of Data Management

Professor
Daniel Vitaver, B.Sc., EMBA

daniel.vitaver@georgebrown.ca

DATA INTEGRATION & INTEROPERABILITY

The DAMA Wheel



INTRODUCTION

- Integration **consolidates data into consistent forms**, physical or virtual
- **Interoperability** is the **ability** for multiple data systems to communicate
- **DII solutions** enable DM functions such as:
 - data migration and conversion
 - data consolidation into hubs or marts
 - integration of vendor packages

DII Management Framework

Definition

Managing the **movement and consolidation of data** within and between data stores, applications and organizations

Principles:

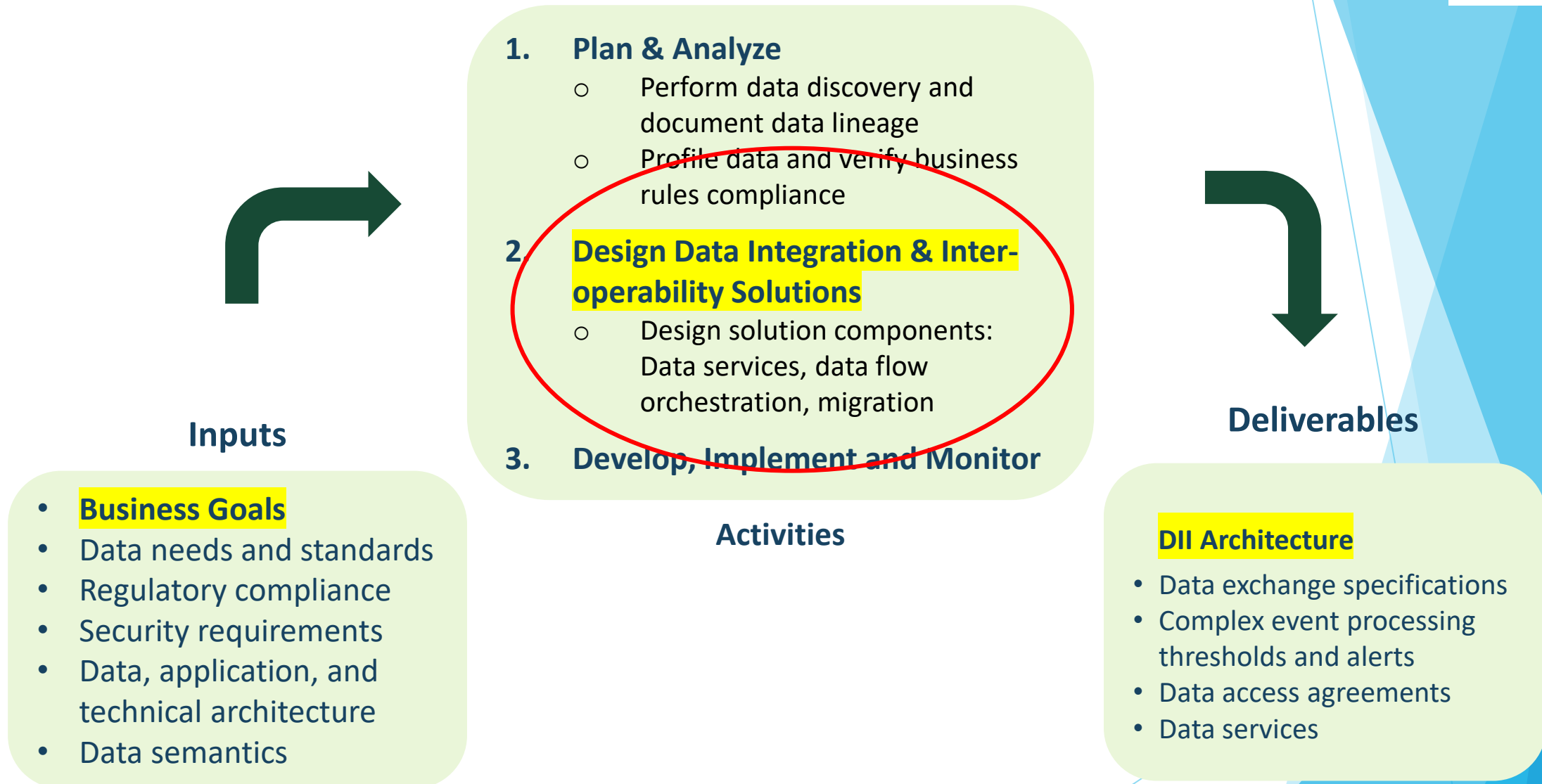
1. Take an **enterprise perspective** in design to ensure future scalability and extensibility
2. Ensure business **accountability**
3. Balance local data **needs** with enterprise data needs

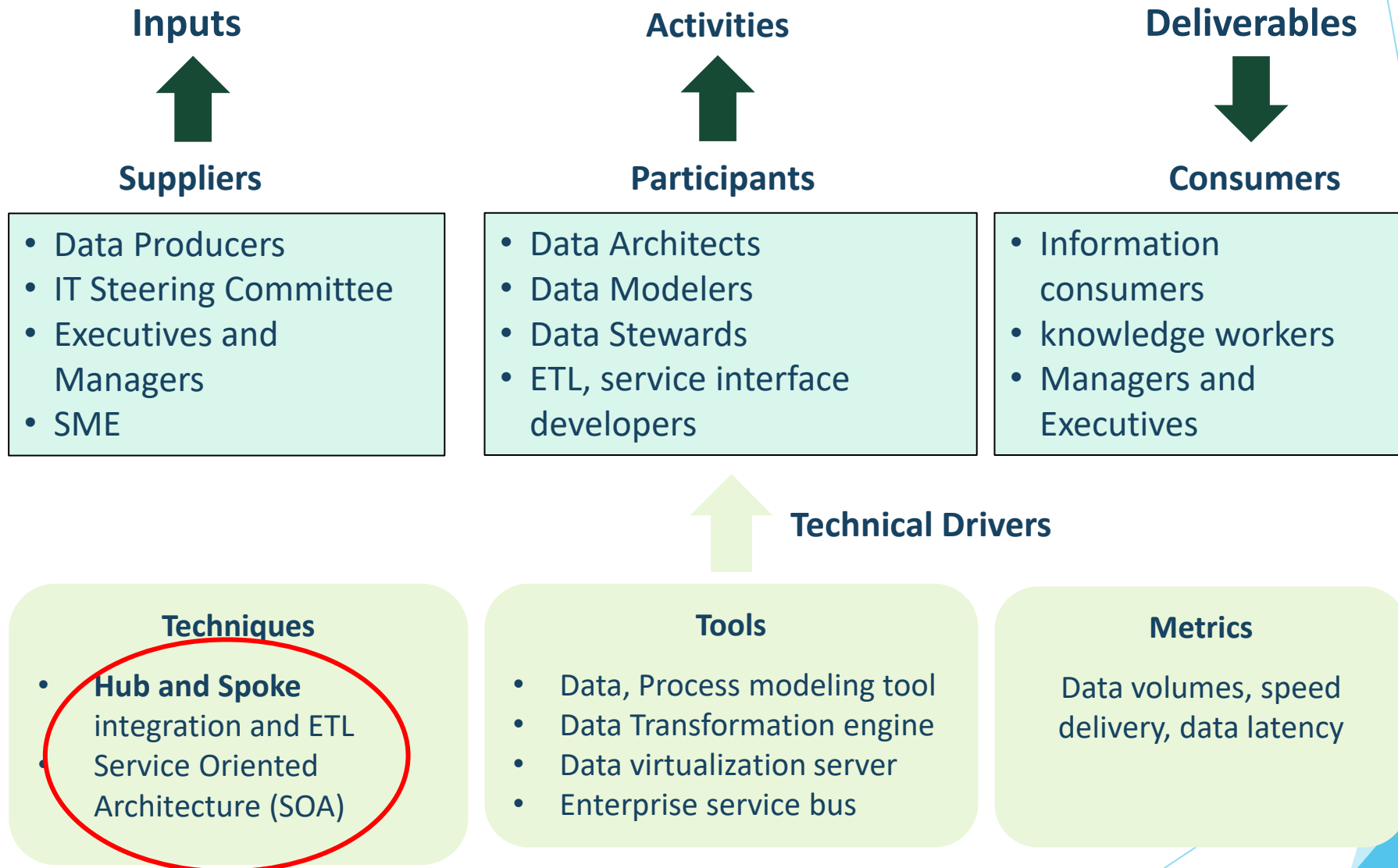
Goals

- **Provide data** securely, regulatory compliance, in the format, timeframe needed
- **Lower cost, complexity**, developing shared models and interfaces
- **Support BI and Analytics**



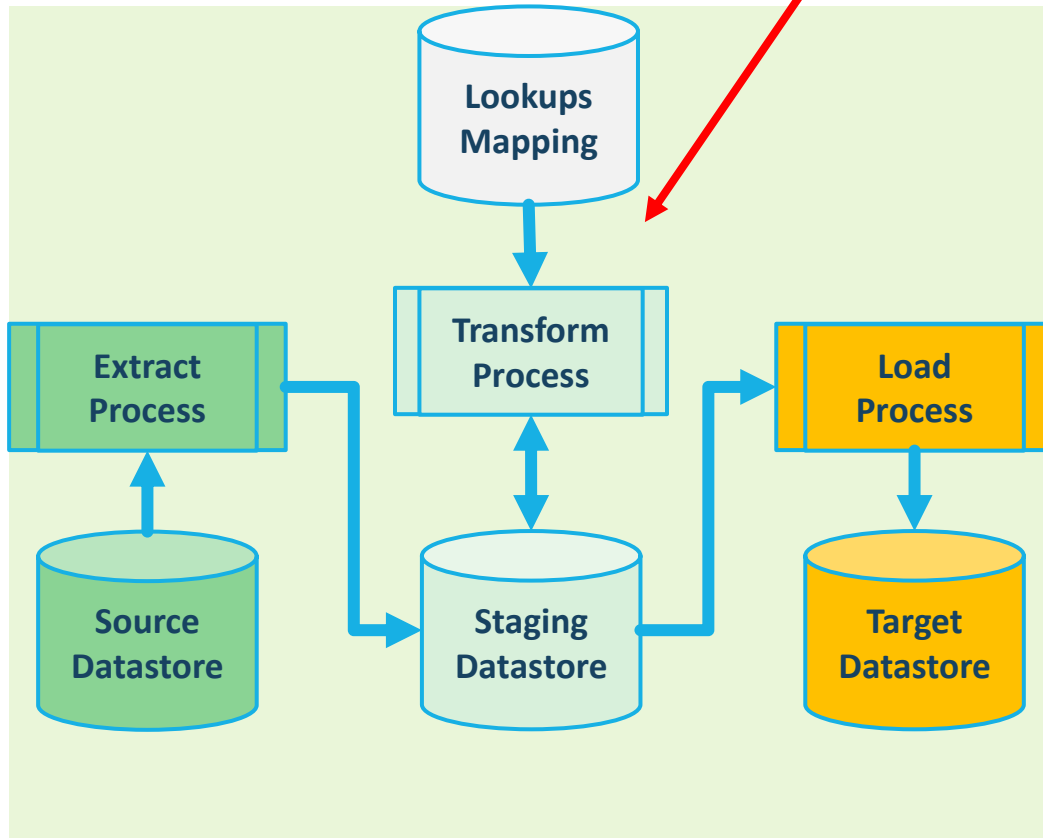
Business Drivers



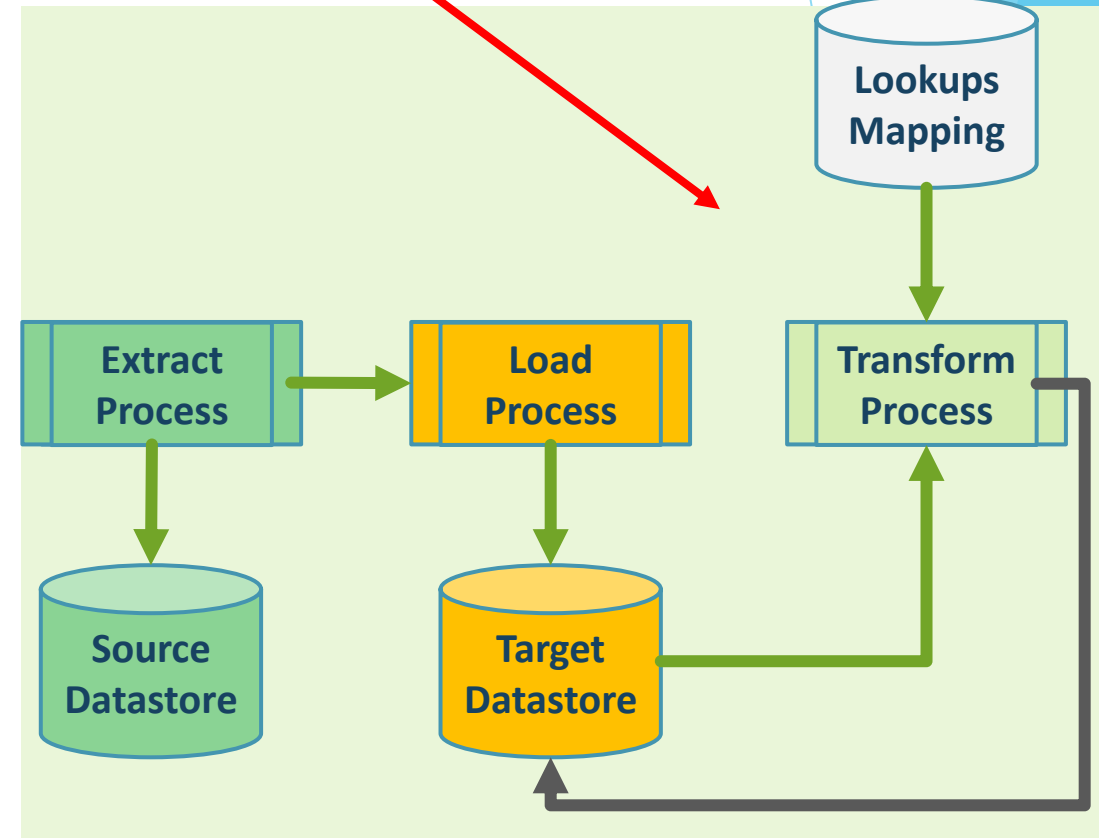


- **ETL – extract, transform, and load**

Transformation System
Formats, Rules, Calculations



ETL



ELT

- **Latency**

It is the **time difference** between when data is **generated** in the source system and when the data is **available** for use in the target system

High

- **Batch:** moves data between applications and organizations in clumps or files either by request or automatically on a periodic schedule

- **Latency – cont'd**

High to low

- **Asynchronous:** system providing data **does not wait** for the receiving system to **acknowledge** update before continuous processing.
 - Either the sending or receiving system could be off-line at a given time without the other system also being off-line
 - Data updates are not immediate, and the delay often is measured in seconds or minutes

- **Latency – cont'd**

- Low**

- **Real-time, synchronous:** no time delay between source and target, data in one dataset perfectly in synch with data in another dataset

- Extremely low**

- **Streaming:** extremely fast data integration solution
 - design to minimize the response time to events
 - streaming data flows from computer systems on a real-time continuous basis immediately

- **Interaction Models**
 1. **Point-to-Point:** systems pass data **directly** to each other.
 - impact to processing due to number of interfaces.
 - data inconsistencies when multiple systems require different data formats
 2. **Hub-and-Spoke:** **consolidates shared data** in a central **hub** that many applications can use
 - consistent views of the data with limited performance impact
 - data warehouse, data marts, master data management system hubs

- **Interaction Models – cont'd**

3. **Publish – Subscribe** model involves systems pushing data out (publish), and other systems pulling data in (subscribe)

- Systems providing data are listed in a **catalog** of data services, and systems looking to consume data **subscribe to those services**
- When data is published, the data is automatically sent to the subscriber

- **Application Coupling (tight or loose coupling)**

To what extent 2 systems are intertwined

- Tight coupling:** 2 systems that have usually a synchronous interface.
It is a riskier operation because if one system fails then both are unavailable
- Loose coupling:** preferred interface design, where data is passed between systems without waiting for a response. It can also be implemented using APIs

- **Orchestration** describes how multiple processes are organized and executed in a system to preserve consistency and continuity
- **Process Control** are the components that ensure shipment, delivery, extraction, and loading of data is accurate and complete

- **Enterprise Services Bus (ESB)**: The bus it is basically a system that acts as an intermediary between other systems, passing messages between them
- **Service-Oriented Architecture (SOA)**: data services are independently provided through well-defined service calls (APIs).

SOA **enables application independence** (loose coupling) and provides the ability to replace systems without major changes to other systems that interact with them

6. GOVERNANCE

- Data Sharing Agreements
- DII and Data Lineage
- Data Integration Metrics

Data Availability

- availability of data requested

Data Volume and Speed

- volumes of data transported
- volumes of data transformed
- volumes of data analyzed
- speed of transmission
- latency between data update and availability
- latency between event and triggered action
- time to availability of new data sources

Solution Costs and Complexity

- cost of developing solutions
- cost of managing solutions
- ease of acquiring new data
- complexity of solutions and operations
- number of systems using data integration solutions

